ENDOMORPHISMS OF AN INJECTIVE HULL OF THE RESIDUE FIELD FORM THE COMPLETION OF THE LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-dualizing/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The description of an injective hull of the residue field as the increasing union of the submodules killed by the powers of the maximal ideal. Invoked inside the counted proof (source lines 1090-1091). Supplies the filtration whose endomorphisms compute the endomorphism ring.
- The identification of the endomorphism ring of each finite length truncation with a quotient of the local ring. Invoked inside the counted proof (source line 1098). Identifies each term of the limit with the corresponding quotient of the local ring.

Source reference. Original lemma lemma-evaluate: sources/stacks-dualizing/pinned-original-source.tex, source lines 942–942. Navigation pointer only: the counted text refers to the Stacks item lemma-evaluate, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-union-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1042–1042. Navigation pointer only: the counted text refers to the Stacks item lemma-union-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let E be an injective hull of κ over R. Then $\mathrm{Hom}_R(E, E)$ is canonically isomorphic to the completion of R.*

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Write $E = \bigcup E_n$ with $E_n = E[\mathfrak{m}^n]$ as in Lemma lemma-union-artinian. Any endomorphism of E preserves this filtration. Hence

$$\mathrm{Hom}_R(E, E) = \lim \mathrm{Hom}_R(E_n, E_n)$$

The lemma follows as $\mathrm{Hom}_R(E_n, E_n) = \mathrm{Hom}_{R/\mathfrak{m}^n}(E_n, E_n) = R/\mathfrak{m}^n$ by Lemma lemma-evaluate. $\square$